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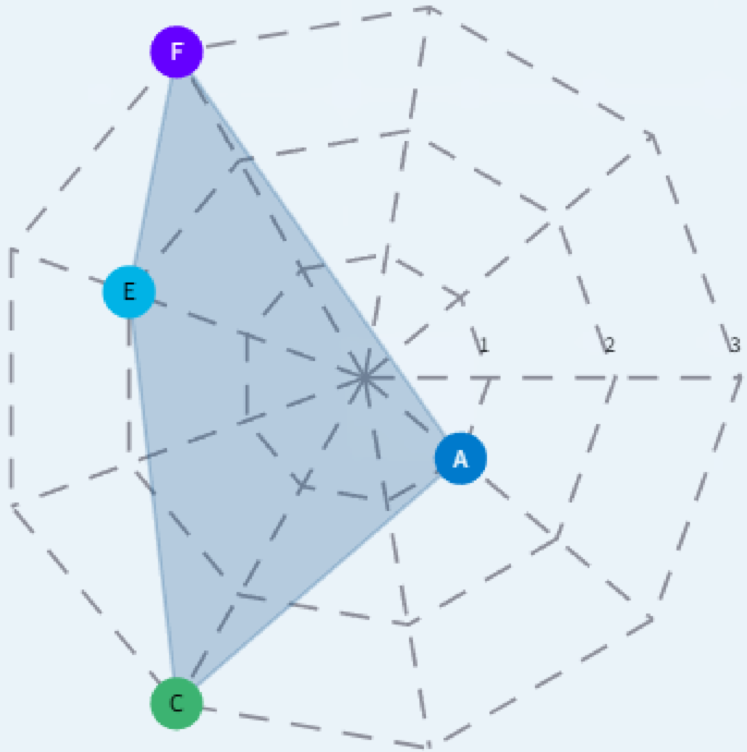
Poll Everywhere

Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



4 0

- A 11% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 33% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 22% Length of Curves
- F 33% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult

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Ex 10.2 Q16 # x, y are func. of t

$x = \sqrt{5-\sqrt{t}}$, $y(t-1) = \sqrt{t}$, $\boxed{t = 4}$

$\boxed{\left.\frac{dy}{dx}\right|_{t=4}}$ ← Aim

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ ① ②

③ $t = 4 \Rightarrow x = ?$

$y = ?$

④ $\left.\frac{dy}{dx}\right|_{t=4} = \cdots$ [Sub $x = ?$, $y = ?$, $t = 4$]

$\frac{dy}{dt}$; $\underline{y(t-1) = \sqrt{t}}$ $\Rightarrow$ Alternative $y = \frac{\sqrt{t}}{t-1}$ [Quotient]

y is a func. of t

$\frac{dy}{dt} = \underline{\underline{y'(t)}}$ $\left[\frac{dy}{dt} = 0 \text{ Wrong}\right]$

Diff. eq. w.r.t. t

$$\frac{d}{dt}(y(t-1)) = \frac{1}{2\sqrt{t}} \Rightarrow \underline{\underline{\frac{dy}{dt}}}(t-1) + y = \frac{1}{2\sqrt{t}}$$

$$\frac{dy}{dt} = \left(\frac{1}{2\sqrt{t}} - y\right) \cdot \frac{1}{t-1}$$

$\frac{dx}{dt}$: $x = \sqrt{5 - \sqrt{t}} \Rightarrow \frac{dx}{dt} = \ldots$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \ldots \quad [\text{sub}]$$

When $t = 4$, $x = \underline{\underline{\sqrt{5 - \sqrt{4}}}}$

$y(4-1) = \sqrt{4} \Rightarrow y = \frac{\sqrt{4}}{3} = \frac{2}{3}$

$$\frac{dy}{dx} = \cdots \qquad (\text{in terms of } x, y, t)$$

$$\left.\begin{aligned} t &= 4 \\ x &= \cdots \\ y &= \cdots \end{aligned}\right\} \text{values}$$

$$\left.\frac{dy}{dx}\right|_{t=4} = \cdots \qquad [\text{sub. values}]$$

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Poll Everywhere

Choose the correct formula for the length of a parametrized curve for $a \leq t \leq b$

.

5 0

$\int_a^b \sqrt{1 + \left(\frac{dy}{dt}\right)^2} dt$

$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dt$

20%

$\int_a^b \sqrt{\underline{\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2}}} dt$

$\frac{d}{dt}\left(\frac{dx}{dt}\right)$

80%

$\frac{dx}{dt} \cdot \frac{dx}{dt}$

$\int_a^b \sqrt{\underline{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}} dt$

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Ex. 10.2 Q28

$x = \frac{1}{3}(2t+3)^{3/2}$, $y = t + \frac{t^2}{2}$

$0 \leq t \leq 3$

$$\text{Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$a = 0$, $b = 3$

$$\frac{dx}{dt} = \frac{1}{\cancel{3}} \cdot \frac{\cancel{3}}{\cancel{2}}(2t+3)^{1/2} \cdot \cancel{2} = (2t+3)^{1/2}$$

$$\frac{dy}{dt} = 1 + t \qquad y = t + \frac{t^2}{2}$$

$$\text{Length} = \int_0^3 \sqrt{\left[(2t+3)^{1/2}\right]^2 + (1+t)^2}\, dt$$

$$= \int_0^3 \sqrt{2t+3 + t^2+2t+1}\, dt$$

$$= \int_0^3 \sqrt{t^2+4t+4}\, dt \qquad \int_0^3 t+2\, dt$$

$$= \int_0^3 \sqrt{(t+2)^2}\, dt = \int_0^3 |t+2|\, dt$$

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Poll Everywhere

Choose the correct graph for $0 \le \theta \le \pi/2$, $1 \le |r| \le 2$

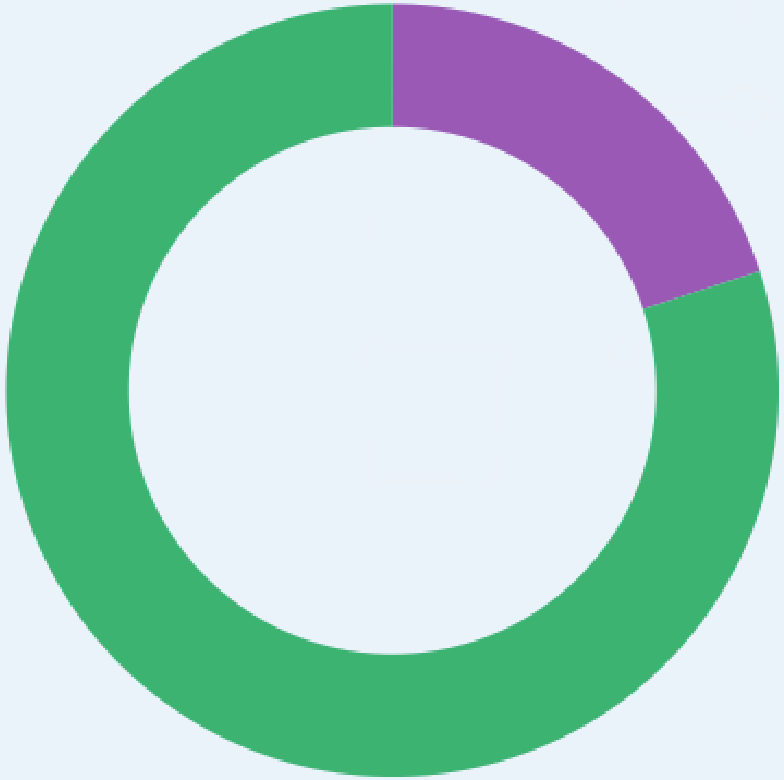
$\Downarrow$

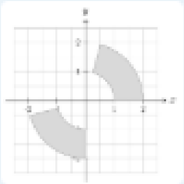
$\boxed{1 \le r \le 2}$

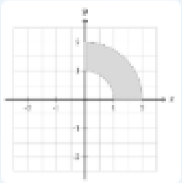
or

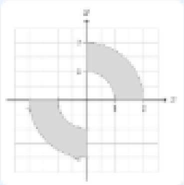
$\underbrace{1 \le -r \le 2}$

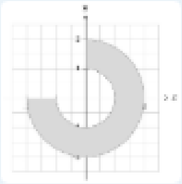
$\boxed{-2 \le r \le -1}$



A 0%


B 20%


C 80%


D 0%


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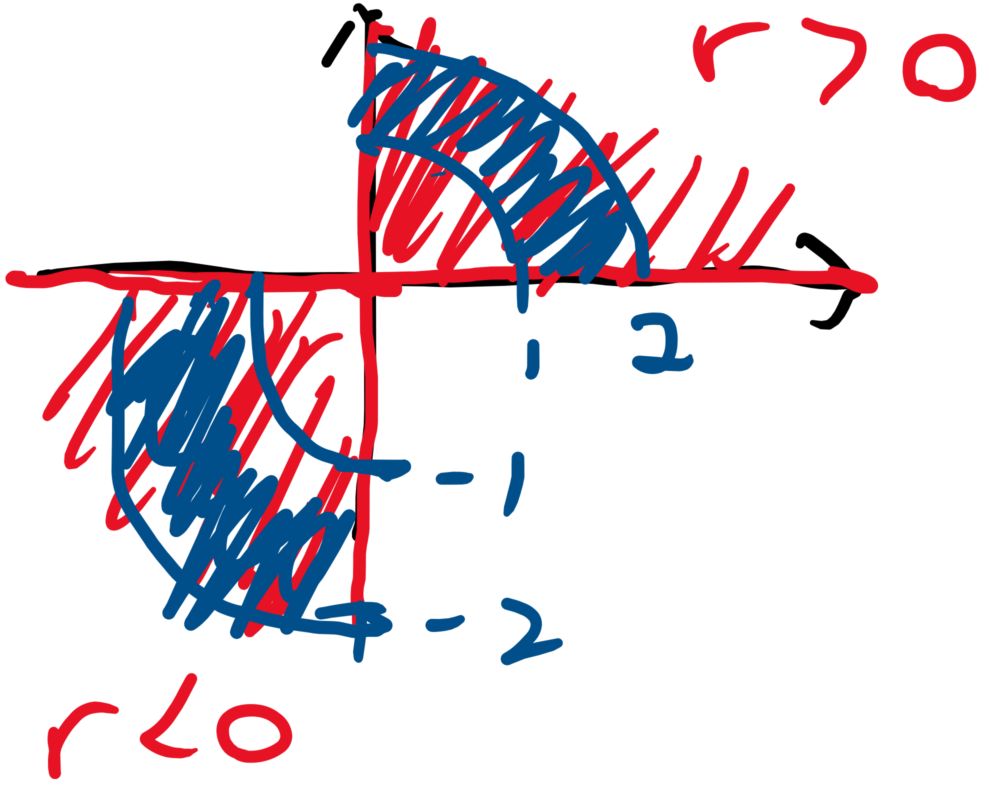
Ex. 10.3
Q26

$0 \leq \theta \leq \pi/2$

$1 \leq r \leq 2 \rightarrow$ pos. ✓

$-2 \leq r \leq -1$



My method to graph polar coordinates:

① Look at $\underline{\underline{\theta}}$ → Graph it

② Identify the regions for $r > 0$ and $r < 0$

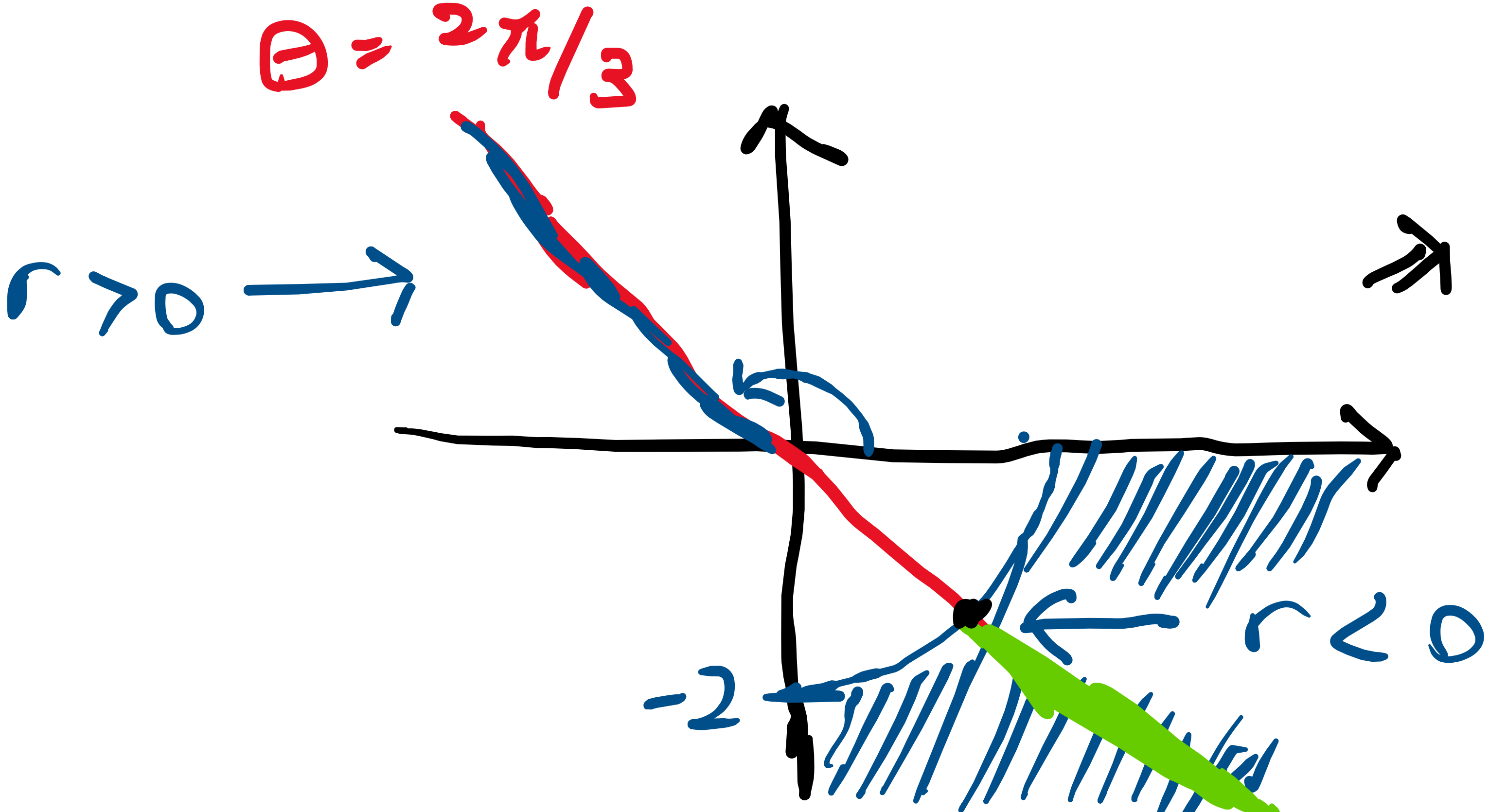
③ Draw circles according to the values of r.

Answer: The intersection of ①, ②, ③

Ex. 10.3 Q16

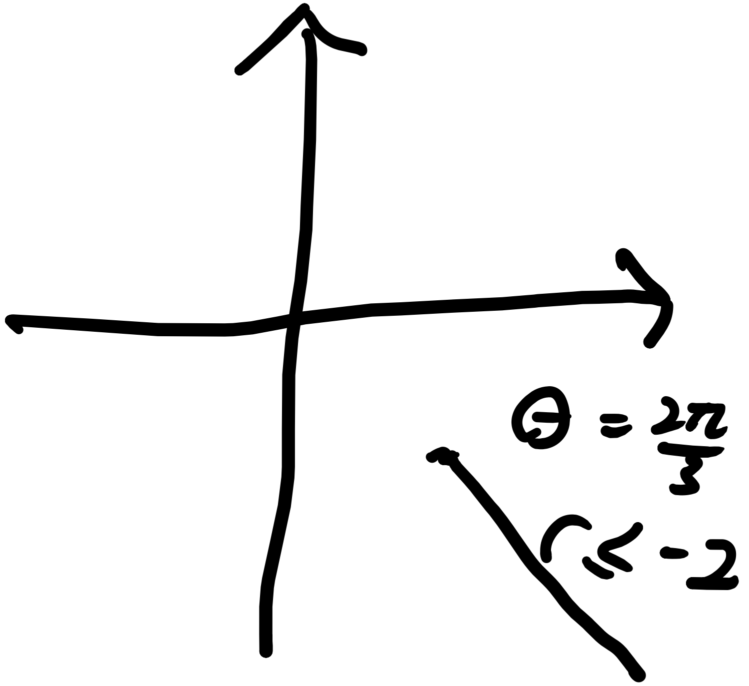
$\theta = 2\pi/3$, $r \le -2$ (negative)

(120 degrees)



$\Rightarrow$

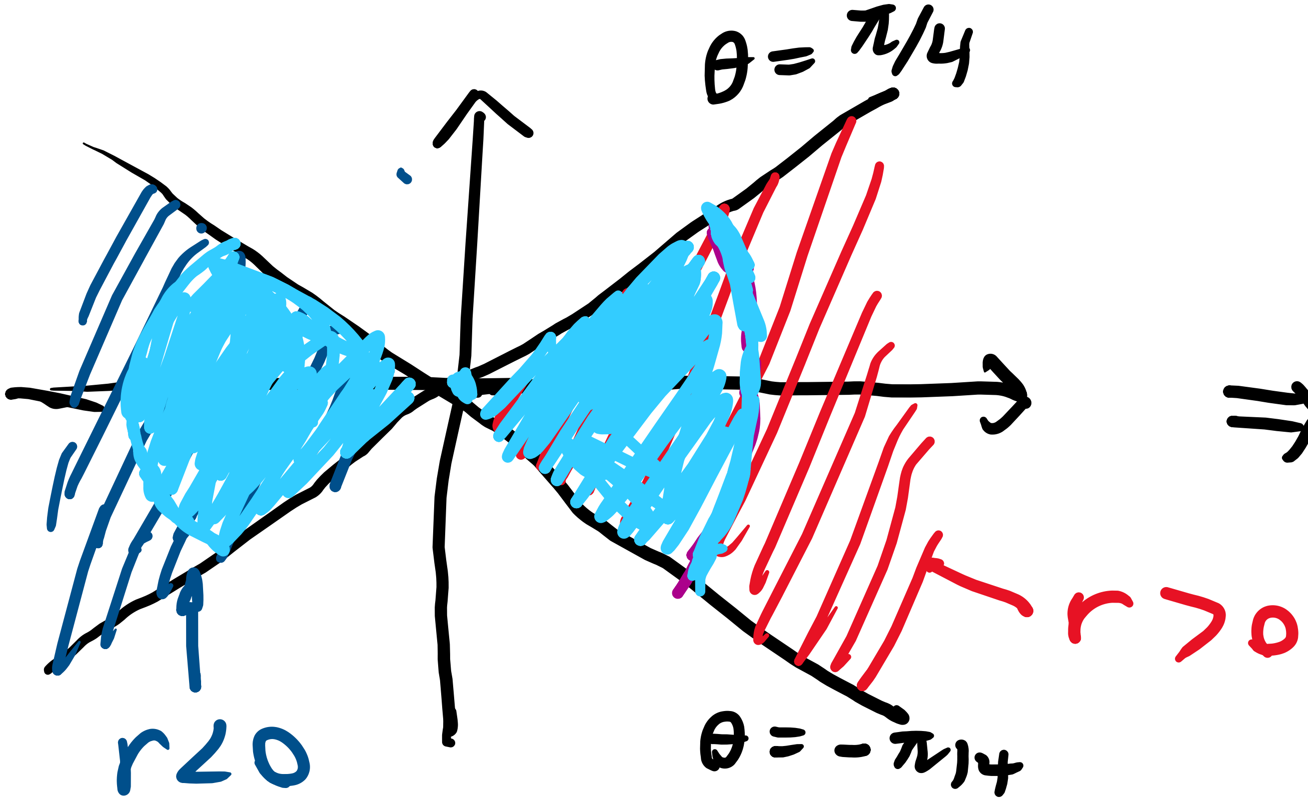
Answer:



Q 24

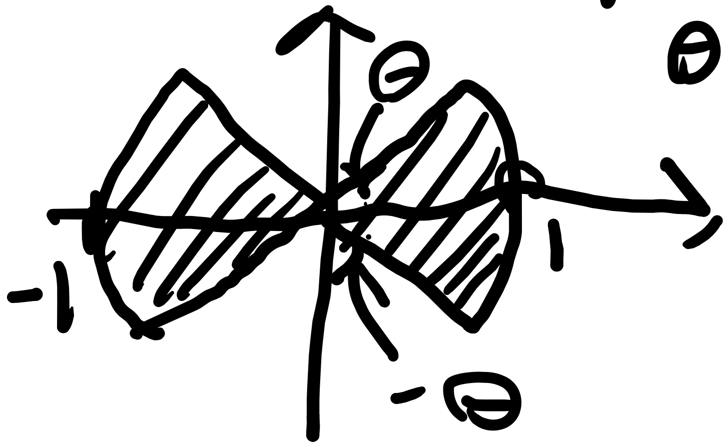
$$\underbrace{-\pi/4 \le \theta \le \pi/4}_{(-45\text{ deg}) \qquad (45\text{ deg.})} \quad , \quad \underbrace{-1 \le r \le 1}$$

$$\underbrace{-1 \le r < 0}_{\text{neg.}} \; ; \; \underbrace{0 \le r \le 1}_{\text{pos.}}$$



$\Rightarrow$

Answer



Specify θ, points

Find Cartesian Eq. from Parametric Eq.

① Direct substitution

$x = \cdots$ in terms of t

$\Downarrow$

$t = \cdots$ in terms of x

sub $t = \cdots$ into $y = \cdots$

② Trigonometric identities

$\sin^2 t + \cos^3 t = 1$ $\qquad$ $\cosh^2 t - \sinh^2 t = 1$

$\tan^2 t + 1 = \sec^2 t$

Ex. 10.1 Q 10

$x = 1 + \sin t$, $y = \cos t - 2$, $0 \le t \le \pi$

$\sin^2 t + \cos^2 t = 1$

$\sin t = x - 1$, $\cos t = y + 2$

$(x-1)^2 + (y+2)^2 = 1$

Graph: $x = \sin t$, $\underline{t = 0.1}$, $t > 0$, x increase

$t = 0 \Rightarrow x = 1,\ y = -1 \Rightarrow (1, -1)$

$t = \pi \Rightarrow x = 1,\ y = -3 \Rightarrow (1, -3)$